

NOVOSIBIRSK OGOURTSOVO, Russian Federation

WMO: 296380

Lat: 54.9108N

Lon: 82.9569E

Elev: 430

StdP: 14.47

Time Zone: 7.00 (E07)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.6	-27.6	-38.0	0.6	-32.5	-32.6	0.9	-27.4	20.4	20.4	18.4	15.8	2.4	210	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	84.3	66.0	81.5	65.0	78.8	64.1	69.1	79.1	67.5	77.1	66.1	75.4	6.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	96.3	73.0	64.1	90.9	72.0	62.5	85.9	70.4	33.6	78.9	32.3	77.2	31.1	75.5	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	15.9	14.1	-35.2	89.0	6.1	3.0	-39.6	91.1	-43.2	92.9	-46.7	94.5	-51.1	96.7
			-35.4	72.0	6.1	1.9	-39.8	73.4	-43.4	74.5	-46.8	75.6	-51.3	77.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.5	-1.2	2.7	21.3	39.5	51.2	63.9	66.8	61.9	51.3	38.5	20.8	6.9
	DBStd	25.87	15.56	13.37	10.27	10.34	9.04	7.71	5.31	6.28	8.13	8.91	13.05	15.26
	HDD50	6905	1589	1324	889	339	98	6	0	1	80	369	875	1335
	HDD65	11002	2054	1744	1354	765	432	110	45	136	416	821	1325	1800
	CDD50	1606	0	0	0	24	136	423	520	370	120	13	0	0
	CDD65	227	0	0	0	0	5	77	100	40	5	0	0	0
	CDH74	1981	0	0	0	7	95	724	743	343	69	0	0	0
	CDH80	407	0	0	0	0	14	159	151	67	16	0	0	0
Wind	WSAvg	6.4	6.4	6.3	7.0	6.9	7.3	5.4	4.8	5.1	5.6	6.7	7.6	7.5
Precipitation	PrecAvg	24.1	1.6	1.1	1.5	1.5	1.9	2.5	3.1	2.2	1.8	2.0	2.7	2.3
	PrecMax	29.8	2.7	2.3	3.2	3.7	4.4	5.0	5.5	3.9	4.1	4.9	5.4	4.6
	PrecMin	18.0	0.4	0.1	0.2	0.2	0.6	1.1	1.0	0.6	0.7	0.9	1.1	0.9
	PrecStd	3.3	0.7	0.5	0.7	0.7	1.0	0.9	1.1	0.8	0.7	0.8	1.0	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	33.8	46.1	73.7	81.7	87.1	88.1	86.5	82.0	66.9	47.3	34.6
		MCWB	31.7	31.2	38.7	54.8	58.0	66.6	67.9	67.2	62.8	53.3	43.4	32.9
	2%	DB	28.9	30.6	40.8	66.2	76.4	84.3	83.7	81.3	74.2	61.1	43.3	32.7
		MCWB	27.3	28.6	34.9	51.8	56.2	65.2	67.1	65.5	58.9	50.0	39.6	31.4
	5%	DB	25.5	27.4	38.1	61.2	72.4	81.5	81.0	77.9	69.1	56.3	38.9	30.8
		MCWB	24.2	25.7	34.0	49.4	55.2	64.6	66.3	64.2	56.6	47.9	36.2	29.5
	10%	DB	21.3	22.8	35.6	55.2	67.5	78.4	78.3	74.4	64.8	51.2	35.6	27.1
		MCWB	19.8	21.0	32.5	46.4	53.0	63.3	65.5	62.3	54.8	45.4	33.4	25.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.8	31.6	38.9	56.5	62.6	70.2	71.7	70.5	63.8	54.3	44.1	33.2
		MCDB	33.9	33.1	45.0	69.4	75.5	82.8	81.4	80.5	76.8	64.0	46.6	34.0
	2%	WB	27.4	28.6	36.0	53.3	58.6	67.8	69.6	67.5	60.2	51.1	39.9	31.9
		MCDB	28.8	30.5	39.4	64.6	71.9	79.3	79.4	76.9	70.5	59.3	42.7	32.7
	5%	WB	24.1	25.6	34.2	50.1	56.6	66.2	68.2	65.9	58.0	48.7	36.4	29.4
		MCDB	25.4	27.4	37.6	59.6	68.9	77.6	77.5	75.5	67.6	55.5	39.0	30.7
	10%	WB	19.9	21.0	32.5	46.5	54.2	64.6	66.9	64.0	55.8	45.7	33.6	25.7
		MCDB	21.2	22.7	35.4	55.2	66.3	75.3	75.3	72.4	63.4	50.9	35.3	27.2
Mean Daily Temperature Range	5% DB	MDBR	14.3	16.2	16.4	17.5	19.1	18.3	17.0	17.9	17.3	12.7	10.3	13.1
		MCDBR	14.1	14.1	16.9	26.3	26.5	23.6	21.6	23.3	25.4	22.4	10.1	10.4
		MCWBR	13.6	13.5	12.8	14.5	12.0	9.3	8.2	9.9	11.8	12.4	8.0	10.5
	5% WB	MCDBR	14.4	14.4	13.8	24.5	22.4	20.2	17.9	20.3	21.9	20.3	10.1	10.6
		MCWBR	14.0	13.9	11.4	14.2	12.2	8.9	8.2	9.3	10.8	11.9	8.1	10.8
Clear-Sky Solar Irradiance	taub	0.294	0.329	0.378	0.409	0.407	0.410	0.437	0.413	0.373	0.355	0.328	0.297	
	taud	2.004	1.958	1.907	2.080	2.211	2.287	2.210	2.262	2.318	2.231	2.040	1.954	
	Ebn at Noon	187	224	241	256	265	266	254	253	248	219	170	151	
	Edh at Noon	23	35	46	45	42	39	42	37	31	26	22	19	
All-Sky Solar Radiation	RadAvg	300	629	1095	1476	1698	1912	1712	1446	976	490	240	193	
	RadStd	26	36	82	91	186	169	112	108	115	66	42	31	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page